PACK WITH LOVE CLUB

From Data to Impact

24-Lesson Project-Based Math Curriculum

Project-Based Math for Social Impact • 2026–2027 • Revised September 6, 2026 (v3)

Annual Challenge: Choose, source, price, market, sell, and evaluate a Pack with Love fundraising product — the product is chosen by the class in Lesson 3; the price is set by the class's own survey data.

Learning Path: Mission → Business → Product & Source → Research → Production → Sales → Analysis → Impact

Core Learning Cycle: Question → Prediction → Data → Math → Decision → Action → New Data → Reflection

Statistics Backbone: Pose a Statistical Question → Decide What to Measure → Collect Data → Summarize → Analyze → Interpret → *Decide & Act* (the course's added step). The class runs this cycle twice: once on the pricing survey (L4–L8, L13) and once on real sales (L18–L21).

Fieldwork (3): Survey Week (homework fieldwork, ≈ Oct 18–25) · Sales Event I · Sales Event II (January). All other sessions are Sunday classes.

Decision Ledger: Fixed by the end of L3 — the product (class recommendation, confirmed by Adult Advisors) and an initial sourcing lean. Decided by data — the price (L13), design details within the product (L9), demand forecast and venue (L16).

Phase 1 · Understand the Business & Choose the Product (3 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
1	What Is a Business? — Fundamentals (Guest: Prof. Kevin Jialin Sun)	What has to be true for a business to exist — and to survive?	Customer · product · price · cost; Revenue – Cost = Profit; cost types (seed); risk	Prof. Sun walks a familiar local business through four boxes — customer, product, price, cost — then the one equation that governs everything after: Revenue (price × sold) – Cost = Profit. Risk defined as "what makes the equation come out wrong." Use of funds and who decides (Charter). Students sketch a business they have actually bought from, with best-guess numbers, and challenge a partner's numbers.	Business Sketch + 2 Business Questions per student
2	How Products Move — Supply Chain, Framework and Reality (Guest: Tony Yao)	How does a product move from an idea to something a customer can actually receive — and what does the textbook not tell you?	Sequence, order of magnitude, MOQ, lead time, cost/time/risk trade-off	Label flip: a "US wholesale" item and a custom sample both read Made in China — domestic vs. import is about who does the importing. Tony walks the full chain on a real order he has run, then the three entry points: upstream (custom in China), downstream (US wholesaler), middle (stock + local value-add). Students map three lanes and mark cost, time, and risk points. Tony answers the Lesson 1 question cards.	Supply Chain Map + Business Risk List
3	Choose the Product, Find the Source — Feasibility Screen & Landed Cost (Guest: Joyce Xie)	Which of these products should we sell — and where in the world would we actually get it from?	Landed-cost estimation, percentages, rates, source reliability	Students shortlist three candidates from real retail field research. Joyce teaches sourcing on a live supplier-platform search (teacher's screen; no student accounts): search terms, reading a listing — price tiers, MOQ, sample cost, lead time, port, weight, verification — and why a listed price is not a quoted price. Teams fill a Sourcing Comparison Sheet, build a landed-cost estimate from their own numbers (product + air/sea freight + tariff + clearance), compare with a US wholesale price, and draft a supplier inquiry. Charter rule stated aloud: students search, compare, draft; adults send, quote, order.	Sourcing Comparison Sheet + Landed-Cost Estimate + Supplier Inquiry Draft + Product Decision Record

Phase 2 · Market Research & Statistics (5 sessions incl. survey-week fieldwork)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
4	Pose the Statistical Question & Design a Survey That Can Be Trusted	The product is decided. What decisions are still open — and how do we ask without pushing people toward our answer?	Statistical question, variables (categorical vs. quantitative), question wording bias, prediction	Introduce the statistics backbone diagram. Contrast a factual question (our landed cost) with a statistical one (what would people pay?). Build the variable list — would you buy (categorical), most you'd pay (quantitative), buy at $5/$7/$10, color/style, grade — and classify each. Sort leading vs. neutral wording. Honest warning: stated willingness-to-pay runs high; January will measure the gap. Each student writes a prediction in context; teams draft an 8–10 item pricing survey.	Prediction Sheet + Survey v1
5	Population, Sample & Bias — Sampling at School	If each of us asks 30–50 people at our own school, whom can the results represent?	Population, sample, simple random sample, stratified sample, sampling bias, fraction	Name random and stratified sampling. Build a school-based sampling plan: each student's school is a stratum; quota cards require mixed grades, genders, and non-friends. Simulate the bias created by asking only friends. Draft the School Support Request (students write the ask; an Adult Advisor submits it).	Sampling Plan + Quota Cards + School Support Request (draft)
6	Survey Launch Lab — fieldwork week	Will strangers answer the way we practiced?	Data collection, coding, tally, interviewer effect	In class: pilot the pricing survey on classmates, practice neutral asking, code and tally pilot data, finalize quota cards. Homework (Survey Week, ≈ Oct 18–25): each student collects responses at their own school under the approved School Support Request, following school rules and the safety note.	Pilot Data + (due next week) Raw Data Set, 30–50 responses per team
7	From Raw Data to Percentages — and to Averages	If 21 of 40 people say they'd pay $7, is that actually a lot?	Frequency, fraction, percentage; dotplot, mean, median, spread; sampling variability (intuition)	Combine every student's school data into one class dataset. Frequency table and percentages for categorical answers; a dotplot of "most you'd pay" with mean, median, and spread for the quantitative one. Split the dataset by school and watch the proportions move — the first look at sampling variability. Check totals and errors.	Clean Data Table + Percentage & Distribution Summary
8	Graphs, Evidence & the Price Signal	How can we make the market message understandable in 10 seconds?	Bar graph, dotplot/histogram, two-way table, comparison	Create graphs; compare schools and grades with a two-way table; use Claim–Evidence–Reasoning to state what the survey says about price and demand. AI checkpoint: let AI draft one chart, then critique what it missed or got wrong.	Market Research Brief + Preliminary Price Signal

Phase 3 · Product, Production & Business Math (6 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
9	Finalize the Product Specification	How do survey preferences become a specification a factory can use?	Measurement, ratio, constraint	With the product fixed since L3, use survey color/style preferences to finalize size, color, packaging, and Pack with Love branding; decide which requirements are non-negotiable versus flexible.	Product Specification Sheet
10	Cost per Unit	What does one product really cost?	Unit rate, fixed/variable cost, estimation	Break down sample, tooling, packaging, production, shipping, and other costs; name fixed vs. variable (seeded in L1); compare different MOQs.	Unit Cost Model

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
11	Compare Suppliers with Evidence	Is the cheapest supplier always the best supplier?	Weighted criteria, score, percentage weights	Build a weighted comparison using quality, price, lead time, MOQ, communication, and risk — with two live quotes on the table, returned from the inquiries the class drafted in L3: a China supplier and a US wholesaler. Jiangang Yao reviews the comparison.	Supplier Comparison Matrix
12	Landed Cost & Import Math	Why is factory price not the same as final cost?	Rates, percent, landed cost, currency awareness	Replace the L3 estimate with real numbers: quoted price, actual freight (air vs. sea), current tariff rate, clearance, packaging → landed cost; compare with domestic wholesale. How far off was our L3 estimate, and why?	Domestic vs. Import Cost Scenario
13	Pricing Scenarios	$5, $7, or $10 — which price is actually better?	Revenue, gross margin, scenario analysis	Use the pricing-survey data (L7–L8) and unit cost to model price/volume scenarios; this is where the data sets the price.	Pricing Recommendation
14	Production Decision & Quality Plan	Are we ready to spend real money and place an order?	Budget, quantity, buffer, percent defect	Set quantity, budget ceiling, quality checks, sample approval, and risk buffer; make a go/no-go recommendation. Adult Advisors place the actual order.	Production Approval Packet

Phase 4 · Marketing & Sales (6 sessions incl. 2 sales events)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
15	Mission-Based Marketing	How can we sell the product without turning the mission into a slogan?	Message testing, comparison	Write and test a value proposition that combines product value with the Pack with Love mission.	Core Message + Sales Pitch
16	Forecast Demand & Choose the Venue	How much should we bring — and where should we even set up the table?	Forecast, ratio, expected value, weighted criteria (reused from L11)	Use survey intent and expected traffic to build low/base/high forecasts per venue (expected revenue = price × P(buy) × traffic). Score candidate venues — school event, church gathering, farmers market — on traffic, audience fit, permission difficulty, and cost. Students draft the outreach ask; Adult Advisors make contact and book.	Sales Forecast + Venue Plan & Outreach Draft
17	Sales System & Event Day Kit	What must we prepare and record on sales day so the event runs — and teaches us something later?	Data fields, conversion rate, average revenue, checklists; experiment design	Design the sales tracker, inventory log, and payment-record rules; assign booth roles. Build the Event Day Kit. Name the experiment: Event I → one adjustment → Event II — and identify what else will differ (venue, day, traffic) and why that limits the conclusions. Cash rule: an adult holds all money.	Sales Tracker + Team Roles + Event Day Kit
18	Sales Event I — fieldwork	Does real customer behavior match what people said in the survey?	Real-time tally, conversion, inventory	Run the first real sale using the Event Day Kit. Record inquiries, buyers, price, inventory change, and feedback.	Sales Data Set A
19	Analyze & Adjust Mid-Campaign	Should we keep the plan, lower price, change the pitch, or move locations?	Percentage change, comparison, prediction error	Compare forecast with actual results and choose one data-based adjustment for Event II.	Mid-Campaign Decision Memo
20	Sales Event II / Campaign Close — fieldwork	What happened after we changed the plan?	Cumulative total, rate, comparison	Run the second sale with the same recording system so the two events can be compared.	Final Sales Data Set

Phase 5 · Analyze, Reflect & Impact (4 sessions)

Lesson	Topic	Driving Question	Math & Thinking	Class Task	Deliverable
21	Post-Sale Statistical Analysis	What did we predict correctly, and where were we wrong?	Difference, percent error, segment analysis, stated vs. revealed preference	Compare survey willingness-to-pay with actual purchase behavior across groups and events; measure the stated-vs-revealed gap named in L4. AI checkpoint: AI may compute; every number it produces must be verified by hand before it enters the report.	Post-Sale Data Dashboard
22	Profit, Net Fundraising & Impact Math	Why is sales revenue not the same as the amount we truly raised?	Revenue, cost, net proceeds, margin	Calculate revenue, direct product costs, approved expenses, and net fundraising; convert net fundraising into the backpack-impact metric on the shared goal.	Financial & Impact Summary
23	Business Review: What Would We Do Differently?	If we did this again, which decision would we change?	Evidence synthesis, trade-off	Review major decisions from product choice (L3) through sales; revisit each student's first-impression pick from L3; distinguish a good decision with a poor outcome from a poor decision with a lucky outcome.	Business Review Report
24	Final Impact Showcase — March 7, 2027	Can we use evidence to explain the entire annual project clearly?	Data storytelling, visual communication	Present the mission, product choice, research, numbers, mistakes, improvements, and final impact to parents, community members, and supporters.	Final Presentation + Student Portfolio

Sourcing & the Student Charter

- Lesson 3 teaches students to find and read supplier listings on a global platform, using the teacher's screen and pre-verified listings; students hold no platform accounts.
- Students draft every supplier inquiry; the lead teacher sends them from the club's email (packwithloveclub@gmail.com), covering both international and Chinese-language platforms, and records replies on the Sourcing Comparison Sheet for L11.
- Consistent with Charter §7 and §9: students search, read, compare, and draft; Adult Advisors contact suppliers, request quotes, approve the supplier (Jiangang Yao reviews before L11 and L14), and place the order. No sample fees, deposits, or quantity commitments are made at the inquiry stage.

Fieldwork Design & School Partnership

Survey Week (≈ Oct 18–25) — distributed fieldwork

- Each student runs the class's pricing survey at their own school under an approved School Support Request — 15 students means samples from up to 15 different school communities, a stronger and more diverse sample than any single-site survey.
- Quota cards (from Lesson 5) require mixed grades, genders, and non-friends — the class's structural defense against friendship bias.
- Safety & ethics note for parents: surveys happen only at school among known peers or in supervised settings, following all school rules; no stranger outreach. Participation in the survey is voluntary and anonymous.

School Support Request — how the club asks a school

- Students draft the request in Lesson 5: who we are, what we are asking (a short voluntary survey during lunch/recess or advisory; later, permission for a small fundraising table at a school event), what we will and won't do (no names collected, no sales pressure, adult supervision).
- An Adult Advisor reviews and submits every request to the teacher, counselor, or administration — consistent with the Student Charter: students draft and recommend; adults make official commitments.
- Each approval is recorded (school, contact, date, scope) by the Project Recorder; fundraising at any school follows that school's own fundraising rules.

Sales Events I & II (January) — designed, not improvised

- Venue Plan (Lesson 16): candidate venues — a school event, a church gathering, a farmers market — scored on a weighted matrix (traffic, audience fit, permission difficulty, cost). Students draft the outreach email/call script; Adult Advisors make contact, handle any fees or forms, and book.
- Event Day Kit (Lesson 17): booth checklist · tally sheet · pitch card · price card · inventory log · customer feedback mini-cards — every student role has a sheet in hand; every transaction becomes a data point.
- The two events form the class's own experiment: Event I → one data-based adjustment → Event II. Students learn to name why it is not a *controlled* experiment (venue, day, and traffic also change) and what that limits.
- Cash rule at every event: a supervising adult holds all money; students record, present, and sell. Net proceeds move through ACO's approved channel.
- Event dates are announced to families at least three weeks ahead; a student who cannot attend receives an alternative real role (pre-event prep or post-event analysis).

AP Statistics Connection

The course's skeleton is the conceptual first half of the AP Statistics curriculum, done for real. Terms are introduced after the experience they name, recorded on a running AP vocabulary card in each Student Portfolio, and never taught as computation: no inference formulas, no hypothesis tests. Two AP habits of mind run throughout — every conclusion written *in context*, and every claim built as claim–evidence–reasoning.

- **Unit 3 · Collecting Data** — L4–L6: statistical question, population/sample, simple random and stratified sampling, convenience and voluntary-response bias, question wording bias. L17–L20: observational vs. experimental, and why the sales experiment is not controlled.
- **Unit 1 · One-Variable Data** — L7: categorical vs. quantitative, frequency tables, dotplot/histogram, shape, center, spread.
- **Unit 2 · Two-Variable Data** — L8, L21: two-way tables, conditional proportions, comparing groups.
- **Unit 4 · Probability** — L5: simulation of friendship bias; L16: expected value in the forecast.
- **Unit 5 · Sampling Variability** (intuition only) — L7: proportions across schools; L21: prediction error.

Why middle school students can do this: the same content is placed at the middle grades by the Common Core (6.SP.1–5, 7.SP.1–8, 8.SP.4) and by the American Statistical Association's GAISE framework (Level B). What this course adds is real data with real consequences, which makes the ideas more concrete, not harder.

Program Parameters

- Recommended grades: Grades 6–8 (2026–27 cohort is primarily Grade 7); elementary students welcome with an accompanying parent (support, not substitution). Every lesson carries an upward extension for advanced learners.
- Core sessions: 24 sessions — 21 Sunday classes (≈60 minutes) + 3 fieldwork activities (Survey Week homework fieldwork; Sales Events I & II). Sundays 5:00 PM Pacific / 8:00 PM Eastern.
- Season: September 13, 2026 – February 28, 2027; Final Impact Showcase March 7, 2027. Approximate milestones: product decision ≈ Sept 27 · Survey Week ≈ Oct 18–25 · pricing data ≈ Nov 1 · sourcing decision ≈ Nov 22 · order ≈ Dec 13 · Sales Events in January. Dates are approximate; the real project comes first.
- Tuition: $30 per session; $720 for the 24-session season. Tuition pays for teaching and is not a donation.
- Class size: 15 students. Roles rotate through Market, Product, Finance, and Sales.
- Participation: Students do not need to travel to Africa. The full annual project is completed locally; the profit from class sales is the team's contribution to the Love & Hope Backpack Project's shared $5,300 goal (net of real product costs), which helps buy 1,000 color-it-yourself backpack sets for children across ten villages in Senegal.
- Final assessment: No traditional exam. Students complete a Student Portfolio, Business Review / Impact Report, and the Final Impact Showcase.
- AI use: AI may draft charts and compute — students must critique, verify, and take responsibility for every number. AI reduces unnecessary work, not necessary thinking.
- Contact: packwithloveclub@gmail.com